Histogram equalization is an interesting mathematical procedure because it maximizes the *entropy* of the image, a measure of how much information is transmitted by a fixed number of bits. The fault with histogram equalization is that it mistakes the shear *number* of pixels at a certain value with the *importance* of the pixels at that value. For example, the truck grill and human intruder are the most prominent features in Fig. 23-13. In spite of this, histogram equalization would almost completely ignore these objects because they contain relatively few pixels. Histogram equalization is quick and easy. Just remember, if it doesn't work well, a manually generated curve will probably do much better.

Warping

One of the problems in photographing a planet's surface is the distortion from the curvature of the spherical shape. For example, suppose you use a telescope to photograph a square region near the center of a planet, as illustrated in Fig. 23-15a. After a few hours, the planet will have rotated on its axis, appearing as in (b). The previously photographed region appears highly distorted because it is curved near the horizon of the planet. Each of the two images contain complete information about the region, just from a different perspective. It is quite common to acquire a photograph such as (a), but really want the image to look like (b), or vice versa. For example, a satellite mapping the surface of a planet may take thousands of images from straight above, as in (a). To make a natural looking picture of the entire planet, such as the image of Venus in Fig. 23-1, each image must be distorted and placed in the proper position. On the other hand, consider a weather satellite looking at a hurricane that is not directly below it. There is no choice but to acquire the image obliquely, as in (b). The image is then converted into how it would appear from above, as in (a).

These spatial transformations are called **warping**. Space photography is the most common use for warping, but there are others. For example, many vacuum tube imaging detectors have various amounts of spatial distortion. This includes night vision cameras used by the military and x-ray detectors used in the medical field. Digital warping (or *dewarping* if you prefer) can be used to correct the inherent distortion in these devices. Special effects artists for motion pictures love to warp images. For example, a technique called **morphing** gradually warps one object into another over a series of frames. This can produces illusions such as a child turning into an adult, or a man turning into a werewolf.

Warping takes the *original image* (a two-dimensional array) and generates a *warped image* (another two-dimensional array). This is done by looping through each pixel in the warped image and asking: What is the proper pixel value that should be placed here? Given the particular row and column being calculated in the warped image, there is a corresponding row and column in the original image. The pixel value from the original image is transferred to the warped image to carry out the algorithm. In the jargon of image processing, the row and column that the pixel *comes from* in the